

PREPARED BY



SYSTEMS-INTEGRATOR PARTNER



PREPARED FOR



U.S. Department of the Interior
Interior Business Center · FPPS

— REQUIREMENTS-FIRST MODERNIZATION OF A NATURAL / ADABAS PAYROLL ESTATE

From 7 million lines of Natural to a **verified requirements baseline** a payroll integrator can implement in Oracle HCM.

Devin reads the legacy estate, extracts every rule with a citation to the source line, separates what must be preserved from what must not be rebuilt, and proves the result with harnesses anyone can rerun on synthetic data. Nothing is translated to a new language. The destination is the HCM; the product is the validated baseline and the proof.

MISSION

Replace interview-driven requirements gathering for FPPS (Software AG Natural 9.x on ADABAS 8.6, z/OS 2.5) with evidence extracted from the code itself — and hand it to the SI as an HCM-ready baseline.

BUILT BY

Cognition AI (Devin, DeepWiki, Devin Review) as AI software-engineering sub-prime, with SMX / Navancio providing payroll SME, ATO, HCM configuration, integrations, and the SI backbone.

STATUS

DEMONSTRATED end to end on a real Natural for AJAX / ADABAS application, entirely on synthetic data.
ROADMAP at FPPS scale on a masked source slice.

60

business rules extracted,
each cited to a `.NSN` file +
line with a confidence score

29 / 59

numbered requirements /
Given-When-Then
acceptance criteria with
HCM-fit ratings

20 of 31

catalogued message codes
never emitted by any
executable path —
candidates not to rebuild

2

multi-user integrity defects in
the booking transaction that
a line-by-line converter
would copy

\$0.00

variance on the clean
synthetic batch; 24
exceptions surfaced on the
deliberately broken one

118

regression, conformance,
and drift tests green in CI;
every generated artifact has
a `--check` gate

All counts on this page are produced by the analyzers and harnesses in the repository and are reproducible with the commands in the package README. Source estate modelled: Sunny Islands Cruise, a public Software AG Natural for AJAX / ADABAS sample (libraries `CRUISE16`, `RDCRUISE`) that structurally mirrors FPPS. No FPPS source or production data was used or is required.

Direct conversion already failed here. The buyer needs requirements, not a translation.

FPPS is **Software AG Natural 9.x** — roughly 7 million lines across 100,000+ modules — on ADABAS 8.6 and z/OS 2.5, driven by ~7,800 **JCL jobs**. It is not COBOL. A prior attempt to push the code through a general-purpose LLM converter produced output nobody could verify. The alternative quoted by a large integrator — interview-driven requirements gathering at roughly \$30M — is slow, expensive, and still depends on institutional memory that is retiring.

WHAT DOES NOT WORK

Line-by-line conversion

Natural source → LLM "translate" → Java / Python / C# →
hope it matches

- Copies every defect faithfully — including race conditions and duplicate-key generation that only show under multi-user load.
- Rebuilds dead, unreachable, and training-only code because a converter cannot tell it apart from the business.
- Produces a second system nobody understands, with no requirements to configure an HCM against.
- Cannot be verified without production data, so the buyer has to trust rather than test.

VS

WHAT THIS PACKAGE DOES

Requirements-first extraction

Natural source → extract + cite → disposition →
requirements baseline → Oracle HCM (SI) → reconcile

- Every rule, edit, field, process, and interface is traced to the exact source line and scored for confidence.
- Integrity behaviour becomes a requirement ("atomic decrement, unique key") that an HCM satisfies natively — not code to port.
- Dead and unmappable logic is surfaced with evidence and routed to SME sign-off, so it is retired deliberately, not by accident.
- Python exists only as the test harness that proves the HCM output equals legacy behaviour, on synthetic data anyone can rerun.

Sample ↔ FPPS analogy

Booking transaction ≈ pay/personnel transaction ·
message-code edits ≈ payroll validation edits · ADABAS
DDMs ≈ personnel-payroll data model · record-hold
integrity ≈ pay-run integrity · CRUISE16 ≈ core Natural
service libraries · RDCRUISE ≈ Web FPPS presentation
layer.

Why synthetic data matters

Every harness, fixture, and reconciliation on this page runs without production access. That is what makes the offer "run our pipeline on your slice and validate the output yourself" credible to a CIO and an ATO board.

About the size figure

Every artifact uses the verified ~7M lines of Natural. A ~63M-line figure circulating in prep material is unverified and is treated as a question for the customer, not as a fact.

Five stages, one chain of evidence: from the source line to the reconciled HCM output.

Each stage produces artifacts the next stage consumes, and every artifact is regenerated from source by a checked-in tool with a drift gate. Nothing on this page is a hand-written claim about the code.

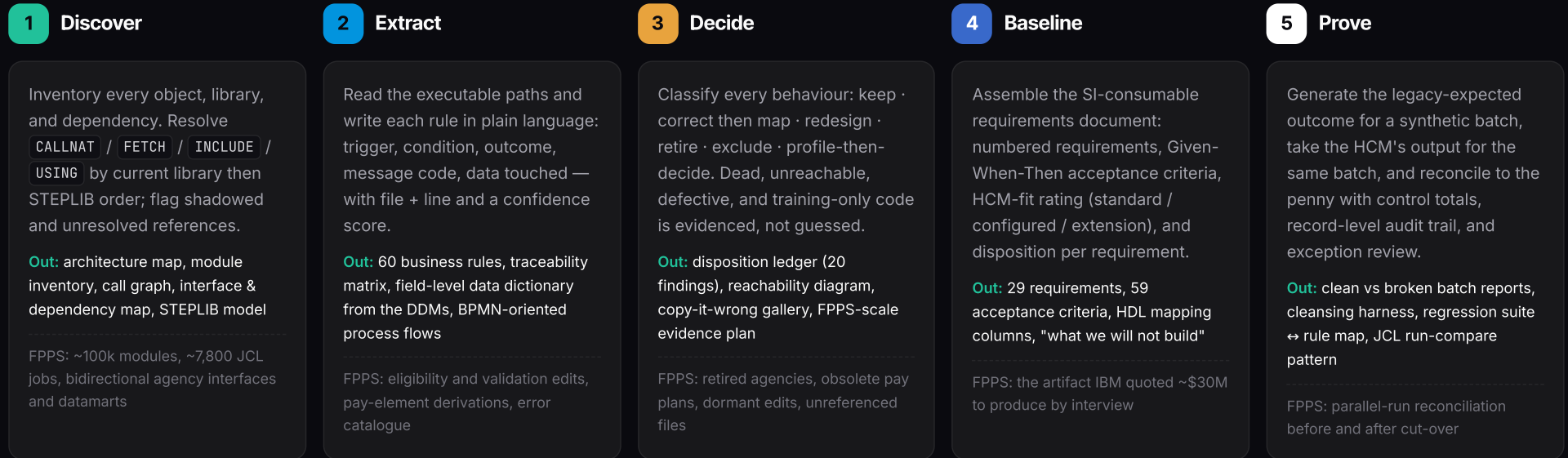
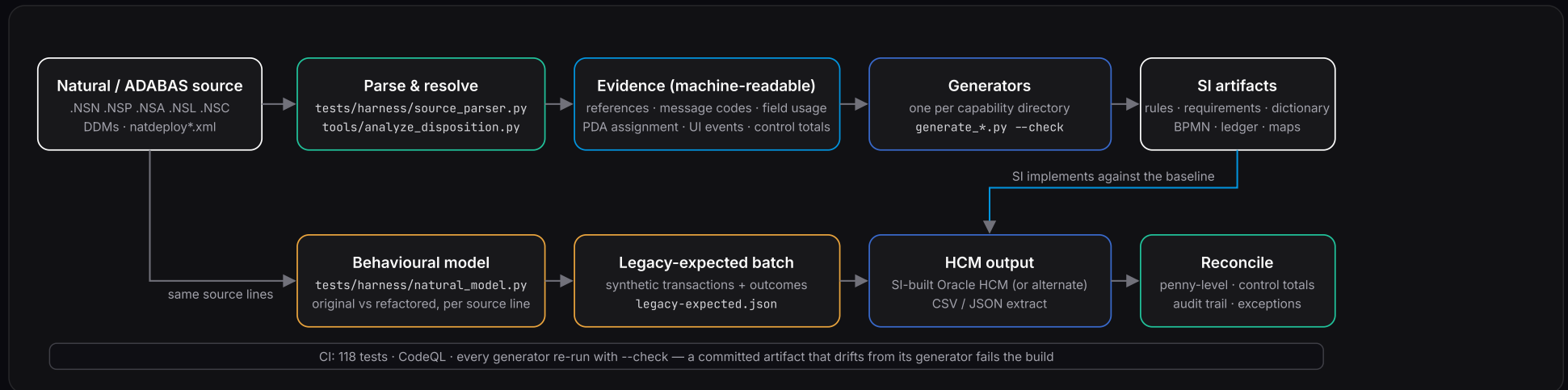


FIGURE 1 Evidence chain — every downstream artifact points back to a source line, and every generated file is regenerated and diffed in CI



Autonomous engineering sessions that read the estate, write the tools, and leave a reproducible trail.

Devin is not a translator. It is an autonomous software engineer that reads the Natural libraries the way a senior developer would — following `CALLNAT` chains, opening the DDMs, tracing a message code back to the edit that raises it — and then writes and tests the analyzers, generators, and harnesses that make each finding reproducible. Human reviewers (Cognition, SMX payroll SMEs, IBC) check the evidence, not a chat transcript.

1 · Comprehend

DeepWiki indexes the repository so anyone can ask "what raises 9918?" and get the answer with the source lines. Devin reads `CRUISE16` (business logic) and `RDCRUISE` (Natural for AJAX presentation), the four DDMs, PDAs/LDAs, copycode, and the `natdeploy*.xml` STEPLIB chains.

2 · Instrument

Devin writes the static analyzer and generators as first-class, tested code. Library-qualified object identity, per-catalogue message-code reconciliation, symbol-owner attribution for field usage, and commented-out statement detection were all added because the real source demanded them — and each fix carries a fixture test.

3 · Parallelise

The ten capability directories were produced by parallel Devin sessions writing into stable, pre-scaffolded targets, then merged and cross-linked. The same fan-out pattern is the plan for FPPS scale **ROADMAP**: partition by library, run in parallel, reconcile at the seams.

4 · Verify

Every generated table has a `--check` gate; the test suite fails if a committed artifact diverges from its generator. Devin Review then reviews Devin's own PR — the review rounds on this package caught parser edge cases (`MOVE BY NAME` , whole-view `RESET` , shadowed same-name objects) before a human ever looked.

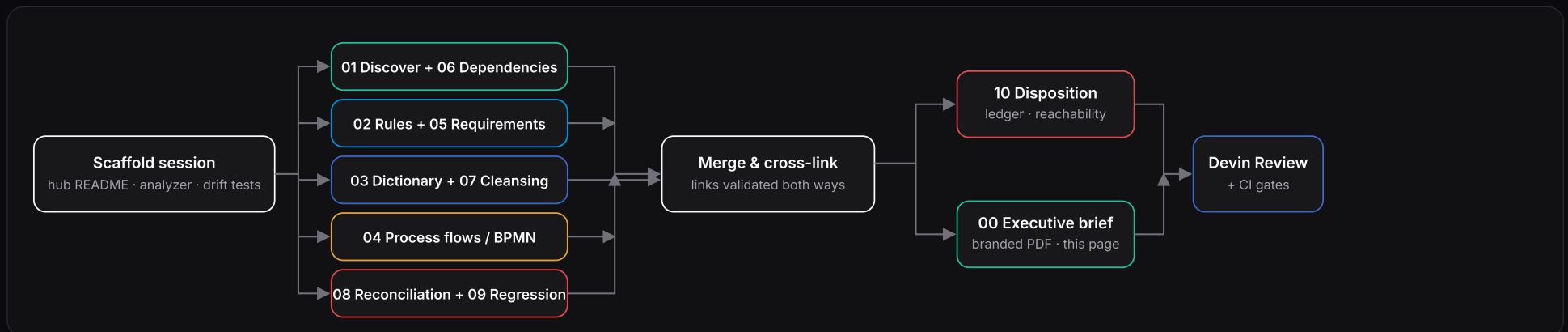
5 · Disposition

Static findings are labelled as candidates with an evidence class and confidence, never as proof. At FPPS scale they are closed with Natural Predict/XRef, runtime traces, profiler data, JCL/scheduler data, ADABAS logs, and SME sign-off — the evidence plan is part of the package.

6 · Hand off

The SI receives requirements, acceptance criteria, an HCM Data Loader-oriented dictionary, BPMN flows, and a reconciliation harness that accepts their HCM's output. Cognition's Python never becomes the system of record; it is the test bench the SI runs against Oracle HCM.

FIGURE 2 Session topology used to build this package — scaffold first, fan out by capability, synthesise last



Eleven package directories, plus the instruments that generate them. Each one answers "how does an integrator consume this into an HCM?"

Everything below is in `fpps-hcm-modernization-deliverable/` on the repository. Maturity labels are applied per artifact: **DEMONSTRATED** runs today and is checked in; **DESIGNED** is specified against the sample; **ROADMAP** needs FPPS inputs this repository does not contain.

00-executive-value-brief

DEMONSTRATED

Executive value brief

Positioning, scope split, proof points, and honest maturity statement for **program leadership**. Markdown master plus Cognition-branded DOCX/PDF.

`executive-value-brief.md` · branded PDF · value-chain figure

01-discoverability-comprehension-baseline

DEMONSTRATED

Discoverability & comprehension

Scope and price what you can see: architecture map, module inventory, and call graph across the business-logic and presentation libraries.

`generate_inventory.py --check` · 31 objects · 62 references · 0 unresolved

02-business-rule-extraction

DEMONSTRATED

Business rules with traceability

Rules read like **payroll eligibility and validation edits**: trigger, condition, outcome, message code, source file + line, confidence.

60 rules · traceability matrix · confidence model · **CONEW-N** decision flow

03-data-model-data-dictionary

DEMONSTRATED

Master-data model & dictionary

The definition of master data that **HCM Data Loader mapping** starts from: formats, lengths, descriptors, usage, lineage, source keys — from the four ADABAS DDMs.

`generate_dictionary_hcm.py --check` · HDL mapping columns

04-business-process-flow-bpmn

DEMONSTRATED

Process flows (BPMN-oriented)

Process definitions an **HCM configuration / BPM team** can ingest and map to Oracle approval and transaction flows; notes on Visio/BPMN import.

list · detail · booking · customer flows · `generate_process_evidence.py`

05-requirements-baseline

DEMONSTRATED

Requirements baseline

The strategic prize: numbered requirements, Given-When-Then acceptance criteria, **HCM-fit rating** and disposition per requirement. Replaces months of interviews.

29 requirements · 59 acceptance criteria · `what-we-will-not-build.md`

06-interface-dependency-mapping

DEMONSTRATED

Interface & dependency map

`CALLNAT / FETCH / INCLUDE / USING` and STEPLIB configuration, plus how the pattern scales to FPPS's **agency interfaces, datamarts, and batch/JCL**.

`generate_dependency_map.py --check` · `natdeploy*.xml` STEPLIB model

07-master-data-cleansing

DEMONSTRATED

Master-data cleansing

Profile → cleanse → reconcile on synthetic dirty master data, **before HDL load** — the leading cause of failed HCM loads addressed up front.

`harness/cleanse_reconcile.py --check` · cleansing rules

08-equivalence-testing-reconciliation

DEMONSTRATED

Equivalence & penny-level reconciliation

Proves **HCM output equals legacy behaviour**: clean vs broken batch, control totals, record-level audit trail, exception review. Validation harness only — never a rewrite target.

`harness/reconcile.py` · `sample-output/{clean,broken}`

09-unit-test-regression-jcl-runcompare

DEMONSTRATED

DESIGNED

Regression & JCL run-compare

Test automation is what the HRD RFI asked for; regression coverage is what **protects a pay run** during and after cut-over. JCL run-compare pattern designed against the sample.

118 tests ↔ rules map · CodeQL CI · `build_suite_map.py --check`

10-migration-disposition-dead-code

DEMONSTRATED

ROADMAP AT SCALE

What not to build

Dead, unreachable, defective, and unmappable logic with **evidence class, confidence, and SME sign-off columns** so nothing is retired by guesswork.

`generate_ledger.py --check` · 20 findings · reachability diagram · evidence plan

tools/ · tests/

DEMONSTRATED

The instruments

The static analyzer, ADABAS simulator, behavioural model, and 118-test suite that make every number on this page reproducible from a clean checkout.

`python3 -m unittest discover -s tests` · `tools/analyze_disposition.py`

The rule a naive converter loses — and the class of bug that silently corrupts a pay run.

The booking transaction `CONEW-N` is the sample's pay/personnel transaction. Its original implementation had two multi-user defects that no single-user unit test and no line-by-line translation would expose. Devin found both by reading the record-hold semantics, proved them with an interleaving simulator, and recorded them as an *integrity requirement* the HCM must satisfy — not as code to port.

ORIGINAL BEHAVIOUR

Read without a hold, then update

Two users read `CRUISE-STATUS = 1` (one place left). Both pass the availability test. Both decrement and store. Result: **two contracts for one place**. Separately, `CONTRACT-ID` was computed as `MAX+1` from a read that did not hold the highest record — two users compute the same key, and `CONTRACT-ID` is a descriptor, not a unique index, so both succeed.

Payroll analog: two personnel actions posting against the same pay period, or two pay records issued under one identifier. It passes every test until the first busy day.

EXTRACTED REQUIREMENT

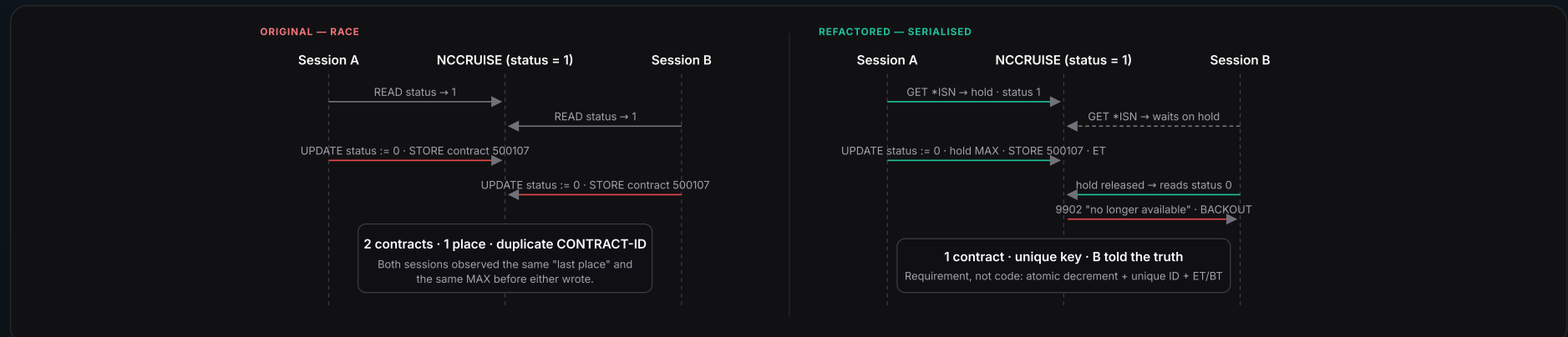
Test-and-set under hold; serialised key generation

```
R1. FIND NCCRUISE WITH CRUISE-ID = ID-CRUISE-IN-N
G1. GET NCCRUISE *ISN(R1.)          /* hold the record */
IF VAL(CRUISE-STATUS) GT 0
  CRUISE-STATUS := CRUISE-STATUS - 1
  UPDATE (G1.)
ELSE MSG-NR := 9902                /* no longer available */

R2. READ (1) NCCONTRACT DESCENDING BY CONTRACT-ID
UPDATE (R2.)                      /* hold MAX before +1 */
NEWCONTRACTID := CONTRACT-ID + 1
STORE NCCONTRACT → END TRANSACTION /* else BACKOUT */
```

In an HCM this is not custom code — it is a native requirement: atomic capacity decrement, unique enforced identifiers, all-or-nothing transaction. The requirement is what the SI configures; the harness is what proves they did.

FIGURE 3 Two sessions interleaved at the vulnerable statement boundary — original vs. refactored, as executed by `tests/test_concurrency.py` against the ADABAS simulator



A converter would have carried both defects into the new platform with perfect fidelity. Requirements-first extraction turns them into two lines of an acceptance criterion the HCM must pass — and the reconciliation harness below is exactly what catches an HCM that fails it (see check `IN-01`, duplicate `CONTRACT_ID 500107`, and `IN-03`, oversold capacity).

The estate is full of things that do not map to an HCM. We find them, prove them, and route them to a decision.

Rebuilding dead code is the most expensive line item in any conversion, and the hardest to see. The disposition analyzer walks every executable path, reconciles message catalogues against the code that emits them, attributes field usage to its owning view, and separates comments from code. Each finding is a *candidate* with an evidence class and confidence; at FPPS scale it is closed with XRef, runtime traces, JCL/scheduler data, ADABAS logs, and SME sign-off.

20 / 31

message codes catalogued but never emitted — plus 10 more emitted only from commented-out lines

7

parameter fields passed to every service and never assigned — including user, password, and language

23

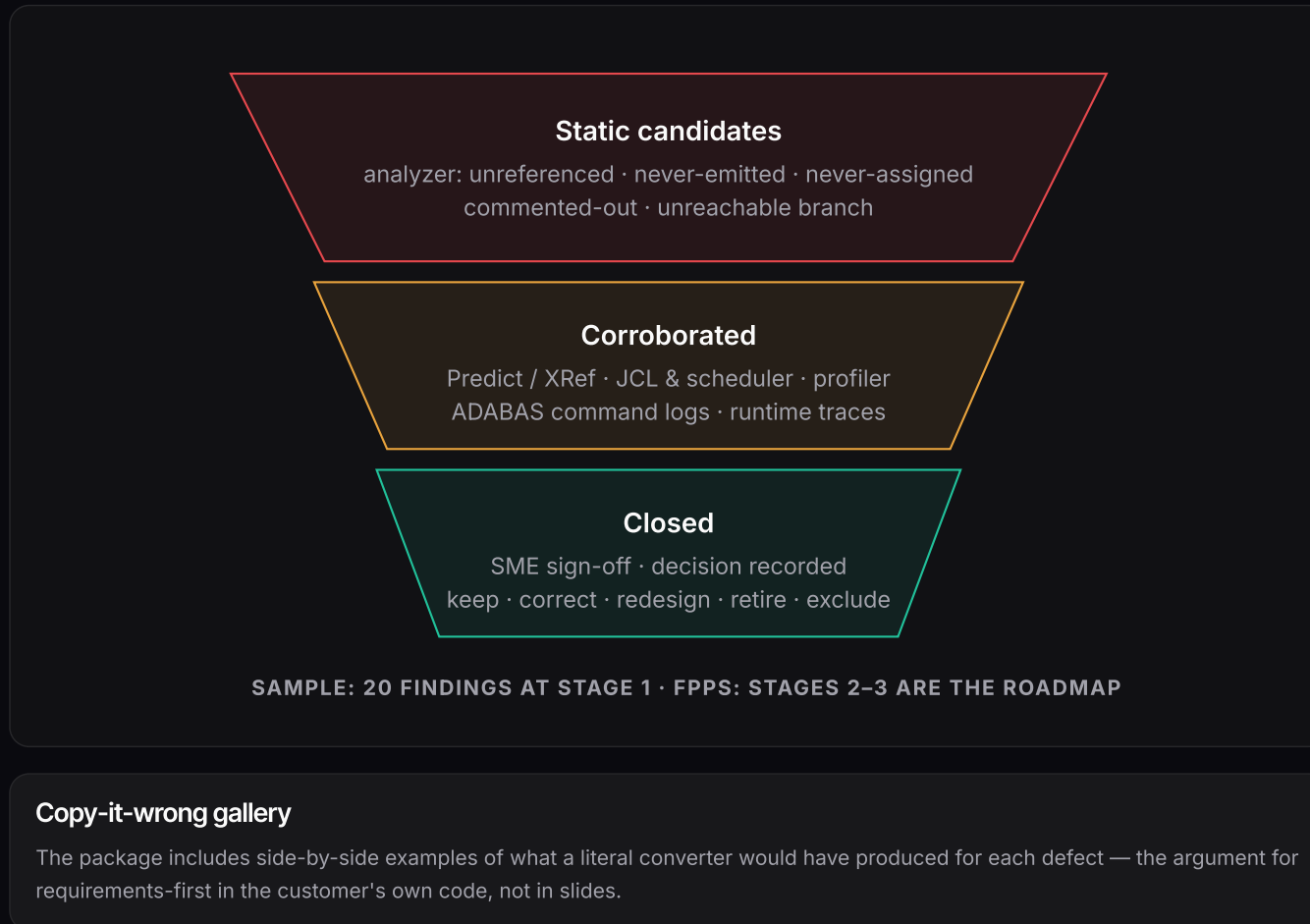
DDM fields never referenced by executable code (every deposit, balance, and cancellation field on the contract)

81

commented-out executable statements, including the entire error-logging copycode and the 2015–2020 year edits

#	FINDING (FROM THE DISPOSITION LEDGER)	WHY IT MATTERS TO THE HCM	DISPOSITION
D-01	<code>P-USER</code> / <code>P-PASSWORD</code> passed to every service, never set or read. "Login" is an identity lookup by ID or e-mail with no authentication.	Do not carry a fake security contract into HCM; use the platform's identity and role model.	DO NOT MAP
D-02	<code>P-LANG</code> never populated, so the German, Spanish, and Portuguese service catalogues are statically unreachable.	Translation scope is a decision, not a requirement inherited from dead branches.	SME DECISION
D-04 / 05	Reservation and booking year edits hard-coded to 2015–2020, and seven other validations — all commented out.	A converter would either reimplement a time bomb or silently drop the edit. Both are wrong; the SI decides.	CANDIDATE ONLY
D-06	Page writes <code>FIRST-NAME-1</code> ; services persist <code>FIRST-NAME-OLD</code> . Typed first names are never saved.	A lineage defect in the person master — correct the requirement, then map.	CORRECT, THEN MAP
D-07	<code>CRGET-N</code> returns 9934 ("modified by another user") for not-found, and the start-value read can show the <i>next</i> record on a miss.	Wrong error semantics propagate into the HCM's message catalogue if copied.	CORRECT, THEN MAP
D-08	23 DDM fields never referenced; 17 not exposed in any view.	Profile the real data before deciding whether an HDL mapping exists at all.	PROFILE, THEN DECIDE
D-09	Stand-alone customer delete by physical ISN, no caller, no audit.	Replace with the governed HCM person-termination function.	REPLACE
D-10 / 11 / 16	Unreferenced subprogram, an image loader with only commented-out call sites, five unused parameter data areas.	Zero HCM requirements; cost only if converted.	RETIRE
D-13 / 14 / 18	Training gate (<code>#STUDENT</code>), exercise fields, hard-coded favourite harbours and cruise IDs in the page.	Classroom and presentation scaffolding, not business behaviour.	EXCLUDE
D-17	Error-logging copycode is 100% comments: runtime errors surface as success.	Record as a platform gap the HCM closes natively — do not "preserve" it.	PLATFORM GAP
D-19	Record-hold test-and-set and serialised key generation in <code>CONEW-N</code> .	The integrity behaviour that must survive as an HCM-native requirement.	KEEP

FIGURE 4 Evidence funnel — static candidates never become "dead" without runtime and SME evidence



Clean batch reconciles to the penny. Broken batch surfaces every planted defect with an audit trail.

Testing is the product. The harness generates the legacy-expected outcome for a synthetic batch of 24 booking transactions from the behavioural model of the Natural source, then reconciles an HCM's output for the same batch: record-level equality, control totals, integrity checks, and rule-outcome checks. The numbers below are the committed outputs of `reconcile.py run-all`.

Clean batch · SYN-2026-09-BATCH-01

RESULT: RECONCILED

24 / 24

records matched

0

exceptions

\$0.00

price variance

CONTROL TOTAL	LEGACY EXPECTED	HCM OUTPUT
Accepted / rejected	11 / 13	11 / 13
Codes 9800 · 9902 · 9904 · 9905 · 9918	11 · 6 · 2 · 2 · 2	11 · 6 · 2 · 2 · 2
Σ price over accepted	14,195.48	14,195.48
Distinct CONTRACT_ID	11 of 11	11 of 11

Broken batch · same transactions

RESULT: EXCEPTIONS (24)

20 / 24

records matched

24

exceptions · 13 checks

+\$2,639.98

price variance

CHECK	EXCEPTION (FROM EXCEPTIONS.CSV)
IN-01	CONTRACT_ID 500107 assigned to two transactions (T-0017, T-0018) — the duplicate-key defect
IN-03	Cruise 196 oversold: 6 accepted against 5 available — the race-condition defect
IN-04	Booking accepted for customer 99999999 who is not in the customer master (should be 9918)
RL-04	T-0016 returned 9800 where legacy returns 9902
CT-04	Σ price differs by +263,998 cents

Record-level audit trail

`audit-trail.csv` records every transaction, the expected and actual outcome, message code, contract ID, and price — the artifact an auditor or an ATO assessor asks for first.

Master-data cleansing, same pattern

Directory 07 runs profile → cleanse → reconcile on synthetic dirty customer master data (formats, duplicates, orphans) so the HCM load starts clean; the same control-total discipline applies.

Reproduce in one command

```
python3 -m unittest discover -s tests -v
python3 tools/analyze_disposition.py
python3 fpps-hcm-modernization-deliverable/08-equivalence-testing-reconciliation/harness/reconcile.py run-all
```

Three parties, one baseline, no trust required.



For DOI Interior Business Center

- **The requirements baseline without the ~\$30M interview programme** — produced from the code, cited to the line, and reproducible.
- A defensible answer to "what are we not rebuilding, and why?" — the disposition ledger with SME sign-off columns.
- Pay-run integrity captured as acceptance criteria, so the HCM is tested for the defects that matter before the first live pay period.
- No production data leaves the boundary during evaluation: everything runs synthetic first, then on a masked slice inside IBC's environment.



For SMX as prime

- An SI-consumable input set on day one: requirements, acceptance criteria, HDL-oriented dictionary, BPMN flows, dependency map.
- A reconciliation harness that accepts *your* HCM output — the parallel-run evidence you need for cut-over sign-off.
- Clear partitioning: SMX / Navancio own payroll SME, ATO, HCM configuration, integrations, and the SI backbone; Cognition owns the AI software-engineering scope as sub-prime (~50–60%).
- The concurrency and dead-code stories answer the "why not just convert it?" objection in the customer's own code.



Cognition Why Cognition

- Devin runs autonomous, parallel engineering sessions — the fan-out a 100k-module estate calls for — and leaves tested tools, not transcripts.
- DeepWiki gives every reviewer the same source-grounded understanding; Devin Review checks Devin's own work before humans do.
- Every number on this page is generated by code that is in the repository and re-run in CI; drift fails the build.
- FedRAMP High authorized boundary (Devin Desktop via Palantir FedStart on AWS GovCloud); DoD IL4/IL5/IL6 compliant.

Next steps

01

Walk the package

Hub README → executive brief → rules, requirements, ledger. Ask DeepWiki anything about the sample estate.

DEMONSTRATED

02

Run the proof

Regenerate the ledger and reconcile the clean and broken batches from a clean checkout — on your machine.

DEMONSTRATED

03

Masked FPPS slice

One Natural library and its DDMs, masked, inside IBC's boundary. Same pipeline, real scale, SME closes the dispositions.

ROADMAP

04

SI implements, harness verifies

SMX configures Oracle HCM against the baseline; the reconciliation harness proves the output equals legacy behaviour, pay period by pay period.

DESIGNED



Cognition AI Platform

The world's first AI software engineer + agentic IDE



×



cognition.ai · Deliverable repository

Maturity labels: **DEMONSTRATED**

DESIGNED

ROADMAP

Cognition AI builds production-grade autonomous software engineering systems, empowering thousands of developers across the largest organizations in the U.S. Government — NASA JPL, the Social Security Administration, and every Department of War branch: U.S. Army, U.S. Navy, U.S. Air Force, U.S. Marine Corps — as well as the defense industrial base, including Palantir, Anduril, Lockheed Martin, and BAE Systems. The platform is security-first by design: FedRAMP High authorized (via Palantir FedStart on AWS GovCloud) and DoD IL4, IL5, and IL6 compliant.

At the core is Devin: cloud-based autonomous software engineering that plans and executes complex, multi-step engineering work across real codebases — autonomous, parallel, and effectively infinite in scale — in production with customers today.

Devin anchors the broader Cognition platform — DeepWiki for codebase intelligence, CLI and IDE developer tools, and Devin Review for lifecycle, CI, and change management — combined with federal audit and compliance automation (NIST, STIG) and Devin Security Swarm: complex, multi-threaded security analysis and remediation built for the hardest problems in the federal market.

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